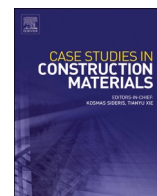




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Case study

Dynamic mechanical properties of lunar regolith based geopolymer: Coupled effects of composition and temperature

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ABSTRACT

This study developed a high-fidelity lunar regolith simulant (GCD-2) that closely replicated the physicochemical properties of lunar samples retrieved by the Chang'e-6 mission. A lunar regolith based geopolymer (LRG) was subsequently synthesized using "one-part" method, in which a dry mixture of GCD-2 (as precursor) and solid sodium silicate (as activator) was combined with water. The dynamic mechanical properties of the LRG were then evaluated using the Split Hopkinson Pressure Bar (SHPB) apparatus. Systematic investigation was conducted to elucidate the influence of precursor Ca/Si and Al/Si ratios in lunar regolith simulant, as well as lunar thermal cycling, on the dynamic compressive strength and energy absorption capacity of the LRG. Two thermal curing regimes were designed to identify the optimal thermal curing window under lunar surface conditions. Microstructural evolution and porosity characteristics were examined using thermogravimetric analysis (TGA), X-ray diffraction (XRD), nitrogen adsorption-desorption (NAD), and scanning electron microscopy (SEM). The results demonstrated that increasing Ca/Si ratio to 0.35 markedly enhanced the degree of geopolymerization, promoting the formation of abundant C-(A)-S-H gel and resulting in a denser matrix with significantly improved dynamic elastic modulus and peak compressive strength. Adjusting Al/Si ratio to 0.50 facilitated the crosslinking of $[AlO_4]$ units within the (C)-N-A-S-H gel network, further improving the dynamic mechanical performance. However, excessive Al/Si ratios hindered geopolymerization kinetics and increased porosity, thereby causing the reduction in the compressive strength of LRG. A synergistic effect between Ca and Al was observed, intensifying geopolymerization and yielding a highly compact microstructure with a peak dynamic compressive strength of 70.77 MPa and superior energy absorption capacity. Furthermore, the optimal time window for mixing and casting LRG on the lunar surface was determined to be within 100 h prior to the high-temperature phase of the lunar daytime. These findings provide both theoretical insights and practical guidance for the development of impact-resistant construction materials to support sustainable extraterrestrial habitat engineering.

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1. Introduction

As the nearest celestial neighbor to Earth, the Moon is widely regarded as a strategic foothold for deep space exploration, due to its unique environmental conditions and abundant natural resources [1,2]. Establishing a permanent lunar base offers immense scientific value, enabling in-depth investigations into the Moon's origin and the evolutionary history of the solar system [3,4]. Accordingly, the lunar infrastructure development has emerged as a central priority of current international space exploration efforts [5]. For instance, the National Aeronautics and Space Administration (NASA) aims to establish a long-term lunar outpost by 2028 to support extended deep space missions [6,7]. At the same time, the China National Space Administration (CNSA) has made significant progress through a series of Chang'e probes and plans to construct a sustainable lunar base by 2030 to enable prolonged exploration activities [8]. However, the transportation of construction materials from Earth remains economically prohibitive [9]. This logistical challenge underscores the critical importance of in-situ resource utilization (ISRU), particularly the use of lunar regolith as a primary construction material [10,11].

Recent studies have demonstrated that different lunar exploration activities demand varying specifications for lunar regolith, with estimated material requirements approximately 13 % for construction applications and 17 % for excavation-related tasks [12]. However, the Apollo missions collectively returned only approximately 380 kg of lunar material, underscoring the extreme scarcity of authentic lunar regolith samples [13]. Consequently, the development of terrestrial substitute materials—lunar regolith simulant—that accurately replicate the geochemical composition, particle morphology, and geological characteristics of genuine lunar soil has become critically important [14]. These simulants played a pivotal role in supporting a wide range of lunar mission scenarios, including drilling operations, landing dynamics, and the testing of in-situ construction materials [1,7,12,15]. The LRSs were generally classified into two categories based on geological provenance at the Moon, highland-type and mare-type simulants [16]. The highland-type simulants were primarily derived from Apollo 16 sample characteristics, including OB-1 [17], LSS-ISAC-1 [18], BH-1 [9], and JLU-H [19]. Among these, BH-1 was specifically engineered for geopolymer synthesis to facilitate lunar construction, while others have been predominantly utilized in particle mechanics studies. The mare-type simulants are generally based on Apollo 14 samples, including JSC-1 [20], FJS-1 [21], CAS-1 [22], TJ-1 [23], HUST-1 [24], and GCD-1 [25]. The simulants specifically developed for geopolymer applications, such as CAS-1, GCD-1, and BH-1, exhibited high fidelity to actual lunar samples in terms of both geochemical and mineralogical composition. In contrast, the simulants tailored for particle mechanics studies, such as FJS-1 and NAO-1, emphasized replication of physical properties over compositional accuracy. Comprehensive evaluations of existing simulants have revealed substantial disparities in physicochemical properties, lunar fidelity, scalability, and cost-effectiveness. This inherent heterogeneity underscores the necessity of establishing application-specific selection criteria and developing tailored simulants to meet the diverse demands of future lunar engineering objectives.

Producing geopolymer binders directly from lunar regolith has emerged as a cost-effective strategy for sustainable lunar construction, with significant progress already reported in recent studies [26–29]. Geopolymer is amorphous aluminosilicate gels formed when silicate- and aluminate-rich precursors dissolve with the alkaline activator, followed by polycondensation into a three-dimensional network structure [30–34]. Due to its rapid setting, high early strength, and excellent long-term mechanical properties, geopolymer becomes a highly promising candidate for lunar construction material [9,29,35,36]. For example, Montes et al. [33] synthesized the JSC-1A based geopolymer, and demonstrated its superior radiation shielding capacity. Additionally, Wang et al. [37] developed a water recovery system that achieved 98.6 % recycling efficiency during geopolymer production, thereby enhancing resource utilization and reducing reliance on external water supplies. Therefore, the lunar regolith based geopolymer (LRG) serves as an eco-friendly and economically viable material, enabling an energy-efficient approach to sustainable lunar habitat construction.

Normally, the mechanical properties of LRG is primarily governed by four primary factors, such as precursor composition, particle size, alkaline activator, and curing environment. First, the strength evolution of LRG is highly correlation to variations in Si, Al, and Ca contents [38]. Accordingly, the geopolymer prepared from lunar regolith simulant representing different lunar regions exhibits composition-dependent mechanical behavior [9]. Second, refining particle size enhances precursor reactivity [39–41]. Alexandri et al. [42] observed that finer particles accelerated reaction kinetics in LRG. Third, deviations from optimal alkalinity can induce incomplete dissolution at low molarity or the formation of deleterious by-products at high molarity levels [29,43–45]. Finally, curing under extraterrestrial conditions introduces additional complexity. Among these, the effects of vacuum curing remained a subject of debate. While Pilehvar et al. [46] and Davis et al. [47] reported adverse outcomes (such as pore coarsening and incomplete geopolymerization), Zhou et al. [41] suggested that reduced carbonation under vacuum conditions may offer certain performance benefits. By contrast, the influence of temperature was more clearly defined. Elevated temperatures have been shown to accelerate geopolymerization, whereas low temperatures substantially hinder strength development [29,41,47–49]. For example, under lunar thermal cycling (−179.8°C to 99.6°C), the porosity of the BH-2 based geopolymer increased by 218 %, accompanied by a 70 % reduction in flexural strength [29]. Similarly, the DNA-1-based geopolymer subjected to thermal cycles ranging from −80°C to 80°C exhibited severe mechanical degradation due to the formation of microcrack networks [50].

In addition to extreme environments, construction materials on the lunar surface must also endure frequent micrometeoroid impacts and seismic activity. During the Apollo missions (1969–1977), seismometers recorded approximately 13000 lunar seismic events [51,52], many of which were attributed to meteoroid impacts and thermal induced stresses [52]. Based on depth and origin, the lunar seismic activity is typically classified into four categories, including deep moonquakes, shallow moonquakes, meteoroid impact-induced quakes, and thermal induced quakes [53]. Among these, shallow moonquakes (occurring at depths shallower than 100 kilometers) are the most intense [54]. Due to their relatively high seismic energy, shallow moonquakes pose the greatest threat to the long-term integrity of lunar infrastructure and are widely regarded as the most seismically potent [55].

2. Research significance

In summary, there are several critical challenges that must be addressed to advance the development of LRG:

1. The moonquakes and meteoroid impacts pose significant threats to the structural stability. However, the existing studies predominantly have focused on the static mechanical behavior of LRG, paying limited attention to their dynamic mechanical properties, including impact resistance. Therefore, to ensure the reliability of the structures constructed by lunar construction materials, it is essential to evaluate their impact resistance under dynamic loading conditions.
2. Recent LRG synthesis efforts have relied on lunar regolith simulants derived from Apollo 14 and 16 sample regions. However, significant compositional heterogeneity in Ca/Si and Al/Si ratios across the lunar surface severely limit the general applicability and diversity of these simulants systems for engineering design. Moreover, the role of Ca-Al compositional interactions in governing mechanical properties measured under dynamic loading remains unclear. Since these compositional ratios critically influence the mechanical response of LRG, a quantitative understanding of the relationships between key oxide components (Ca, Al, and Si) and dynamic mechanical properties is essential for developing structurally robust materials tailored to diverse lunar terrains.
3. The strength development of LRG materials for in-situ lunar construction is subject to stringent time constraints. A lunar daytime lasts approximately 12 Earth days, during which LRG materials must develop sufficient early strength to survive the subsequent cryogenic conditions of the lunar nighttime. Therefore, achieving adequate initial strength within this limited processing window is critical for mitigating cryogenic-induced structural degradation. However, the optimal thermal curing window for LRG remains poorly understood.

To address these challenges, this study systematically investigated the dynamic mechanical properties of LRG with the aim of instructing the design of impact-resistant construction materials for future lunar bases. A novel lunar regolith simulant (GCD-2) was developed, which had geochemical and mineralogical fidelity to the CE-6 lunar regolith. After that, the LRG was synthesized using the "one-part" method, employing GCD-2 as the precursor and solid sodium silicate (SS) as the alkaline activator. The dynamic response of the LRGs was evaluated using a Split Hopkinson Pressure Bar (SHPB) system with varied impact gas pressures. The effects of Ca/Si and Al/Si ratios and thermal cycling on the dynamic strength and energy-absorption capacity of the LRG was then discussed. Additionally, this study designed two batches of samples subjected to different thermal cycling conditions to identify the optimal thermal curing window for LRG. Microstructural evolution was examined through thermogravimetric analysis (TG), X-ray diffraction (XRD), nitrogen

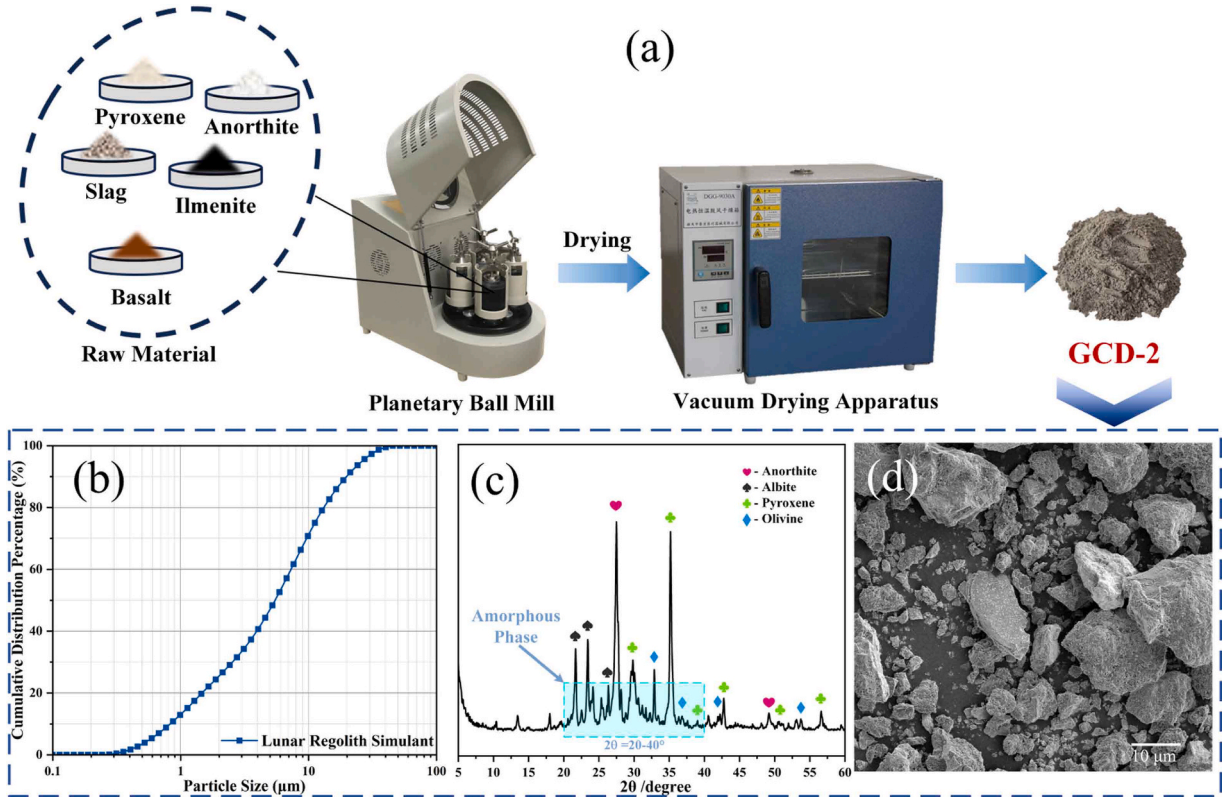


Fig. 1. Characterization of the GCD-2 developed in this study: (a) Preparation Process; (b) Particle size distribution; (c) XRD spectrogram; (d) Microscopic morphology.

adsorption and desorption (NAD), and scanning electron microscopy (SEM). Thus, the influence of compositional heterogeneity on the relationship between macroscopic dynamic behavior and microstructure under extreme curing temperatures can be elucidated. The resulting insights provided a theoretical and practical basis for designing durable, impact-resistant construction materials to support sustainable extraterrestrial habitat engineering.

3. Materials and methods

3.1. Materials

To develop the lunar regolith simulant (GCD-2) that accurately replicated the geochemical and mineralogical characteristics of the CE-6 samples, this study designed a compositionally matched material through precise formulation. The raw materials used for the GCD-2, including basalt, anorthite, pyroxene, ilmenite, were sourced from the Taihang Mountain region in Hebei Province, China. The preparation procedure is outlined in Fig. 1(a). The constituent minerals were weighed to the target ratios, placed in a polyurethane milling jar with 100 mL ethanol, and mechanically activated with 5 mm ZrO₂ balls at 400 r/min for 2 h at 25 °C. The resulting slurry was dried at 105 °C for 4 h to obtain the simulant powder. The subsequent mechanical sieving process was employed to optimize particle reactivity for geopolymer synthesis [41]. Laser diffraction analysis yielded a median particle diameter (D_{50}) of 6.21 μm as shown in Fig. 1(b). As summarized in Table 1, the GCD-2's major oxide contents closely matched those of the CE-6 regolith, with deviations of 0.95 wt% for SiO₂, 1.14 wt% for Al₂O₃, and 0.12 wt% for CaO. As illustrated in Fig. 1(c), the result of X-ray diffraction confirmed that plagioclase (anorthite and albite) was the dominant phase, followed by pyroxene and minor olivine, in agreement with CE-6 mineralogy [56]. A representative microscopic morphology of the GCD-2 can be found in Fig. 1(d). Solid SS granules (modulus = 1.0, containing Na₂O = 25 % and SiO₂ = 50 %, Henan Province, China) were used as the alkaline activator. To adjust the calcium and aluminum contents prior to geopolymer synthesis, analytical-grade (> 99 % purity) CaO and Al₂O₃ powders (Xilong Scientific Co., Ltd., China) were added to the precursor (GCD-2).

3.2. Samples preparation and simulated curing conditions

In this study, a "one-part" method was adopted for the preparation of LRG. This approach involved directly mixing the GCD-2 with the solid alkaline activator, followed by the addition of water. Compared to the conventional "two-part" method, the "one-part" technique had distinct advantages [34,62]. Notably, it eliminated the need for a pre-dissolution of the alkaline activator. This simplification was particularly advantageous in extraterrestrial construction scenarios, as it avoided the risk of leakage associated with the transport and handling of corrosive liquid reagents. The integrated process not only reduced production complexity but also lowered manufacturing costs and energy consumption [30,63]. Furthermore, the "one-part" method significantly enhanced operational feasibility by streamlining the synthesis workflow. As a result, this technique rendered LRG especially well-suited for in-situ lunar construction applications.

In the SHPB tests, the LRG samples with a diameter of 40 mm and a height of 20 mm were fabricated using the silicone mold. The preparation procedure is illustrated in Fig. 2(a). First, predetermined amounts of Al₂O₃ and CaO were added to the GCD-2 to adjust the target Ca/Si and Al/Si ratios. The modified precursor was then mixed with solid SS, followed by the addition of deionized water. The mixture was stirred for 3 min to obtain a homogeneous LRG slurry. After that, the slurry was poured into a silicone mold that had been pre-lubricated with petroleum jelly, and then vibrated on a shaker table for 30 s to eliminate entrapped air. Finally, the demolded samples were placed into the curing apparatus. To ensure consistency in curing onset across all samples, the entire batch preparation was completed within 8 h. Based on previous optimization studies [25], to achieve the workability of the LRG slurry, the formulation in this study were maintained with an alkali activator to precursor mass ratio of 0.18 (solid SS to GCD-2), and a water to binder ratio of 0.25.

To simulate the extreme thermal cycling conditions of the lunar surface, a temperature-controlled simulation system was employed. The curing apparatus (Cryogenic Science & Technology Co. Ltd, China, Fig. 2(a)) enabled precise thermal regulation with an accuracy of $\pm 0.5^\circ\text{C}$. Its temperature output closely followed the lunar surface thermal profile recorded in Fig. 2(b). The curing

Table 1

Comparison of geochemical compositions between the developed GCD-2 and authentic lunar regolith samples, including their corresponding Al/Si and Ca/Si ratios.

	Geochemical composition (wt%)								Ref.	Al/Si	Ca/Si
	SiO ₂	Al ₂ O ₃	CaO	Fe ₂ O ₃	MgO	TiO ₂	K ₂ O	Total			
GCD-2	44.65	15.44	11.78	19.8	3.34	1.74	1.55	98.3	This study	0.34	0.26
CE-6	45.6	14.3	11.9	17.3	7.08	2.7	0.07	98.95	[56]	0.31	0.26
CE-5	42.2	10.8	11.0	22.5	6.48	5.0	0.19	98.17	[57]	0.26	0.26
Apollo 11	42.17	13.78	12.12	15.76	8.17	7.67	0.15	99.82	[58]	0.32	0.28
Apollo 12	46.17	13.71	10.55	15.41	9.91	3.07	0.08	98.9	[59]	0.30	0.23
Apollo 14	48.1	17.4	10.7	10.4	9.4	1.7	0.55	98.25	[25]	0.36	0.22
Apollo 15	46.8	14.6	10.8	14.3	11.5	1.4	0.21	99.61	[9]	0.31	0.23
Apollo 16	45.0	27.3	15.7	5.1	5.7	0.54	0.14	99.48	[60]	0.59	0.35
Apollo 17	42.2	15.7	11.5	12.4	10.3	5.1	0.1	97.3	[61]	0.37	0.27

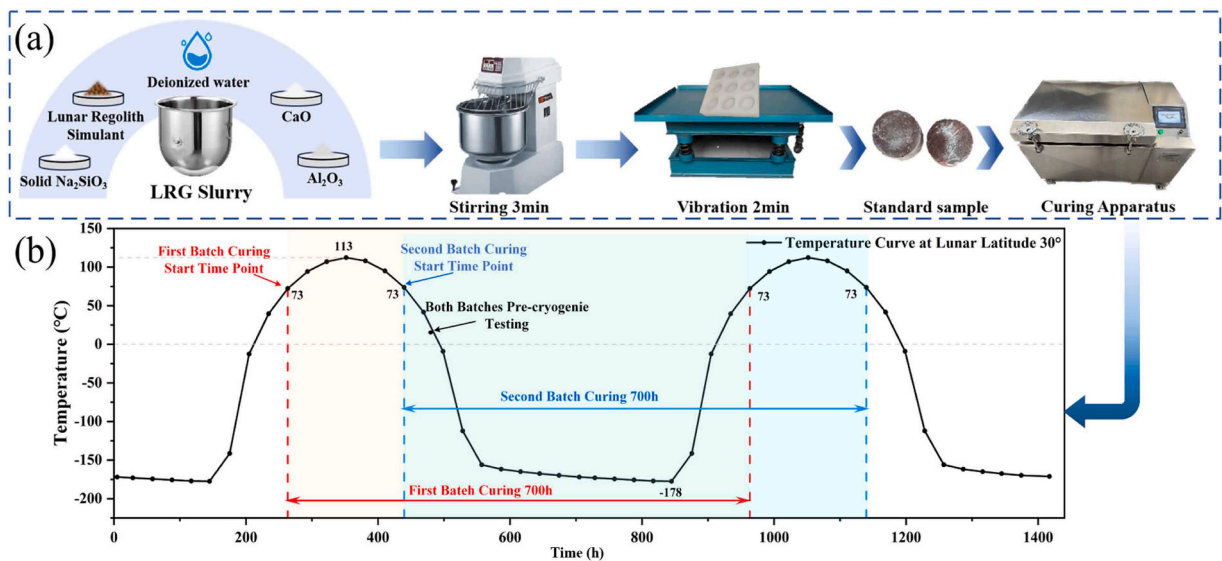


Fig. 2. Preparation and thermal curing conditions of the LRG samples: (a) Preparation procedure; (b) Simulated temperature variation curve at lunar 30° latitude.

apparatus operated within a temperature range of -190°C to 200°C , with both heating and cooling rates and holding accuracy controlled to within 1°C , meeting the high-precision requirements for thermal cycling experiments. The thermal profiles in this study were derived from data provided by the Diviner Lunar Radiometer Experiment onboard NASA's Lunar Reconnaissance Orbiter (LRO) [64], using the surface temperature cycle at 30° lunar latitude as a reference.

To investigate the effects of different thermal curing histories on the dynamic mechanical properties of LRG, this study established two distinct curing protocols as shown in Fig. 2(b). From the figure, it can be seen that the temperature profile performed fluctuations over two complete lunar day-night cycles at a latitude of 30° of the Moon, where daytime surface temperatures peak at 113°C and nighttime temperatures drop to -178°C . To harness the natural high-temperature phase during the lunar day, the first batch of samples began curing at the onset of the heating phase (73°C), subsequently undergoing a full thermal cycle ($73^{\circ}\text{C} \rightarrow 113^{\circ}\text{C} \rightarrow -178^{\circ}\text{C} \rightarrow 73^{\circ}\text{C}$) over approximately 700 h. In contrast, the second batch started during the cooling phase at 73°C and experienced an alternative sequence over 700 h ($73^{\circ}\text{C} \rightarrow -178^{\circ}\text{C} \rightarrow 113^{\circ}\text{C} \rightarrow 73^{\circ}\text{C}$). Although both batches shared similar initial and final temperatures, their curing trajectories diverged significantly. The first batch commenced under relatively high-temperature conditions, whereas the second entered the cryogenic lunar night phase shortly after curing began. This early exposure to low temperatures and shortened high-temperature duration may have impeded the initial geopolymerization reactions. This experimental design aimed to identify the optimal thermal curing window for LRG and to elucidate the influence mechanisms of extreme lunar temperature cycles on their dynamic behavior.

3.3. Experimental procedures

Previous studies indicated that the mechanical properties of geopolymer mainly depended on the Si/Al/Ca compositions of the aluminosilicate precursor and the curing conditions [65,66]. To address the cost-efficiency and diversity demands of ISRU for lunar construction, this study systematically investigated the composition-property relationships within the spatial heterogeneity of geochemical parameters in authentic lunar regolith samples ($\text{Al/Si}=0.30\text{--}0.59$, $\text{Ca/Si}=0.22\text{--}0.35$, as shown in Table 1) from the Chang'e and Apollo missions. In this study, the selected ranges for Ca/Si and Al/Si ratios ($\text{Ca/Si}=0.26\text{--}0.35$, $\text{Al/Si}=0.34\text{--}0.58$) and the incremental adjustments ($\Delta\text{Ca/Si}=0.03$, $\Delta\text{Al/Si}=0.04$) were strictly based on the statistical distribution of measured lunar chemical

Table 2

The varied Al/Si and Ca/Si ratios in precursor for LRG.

LRG Groups	Al/Si in precursor	Ca/Si in precursor	Impact gas pressure	Curing batches
BG	0.34	0.26	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
C1	0.34	0.29	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
C2	0.34	0.32	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
C3	0.34	0.35	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
A1	0.42	0.26	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
A2	0.50	0.26	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
A3	0.58	0.26	0.2 MPa 0.3 MPa 0.4 MPa	First, Second
C3A3	0.58	0.35	0.2 MPa 0.3 MPa 0.4 MPa	First, Second

compositions. These values are within the Ca/Si and Al/Si ranges documented for lunar regolith in Table 1, ensuring that the simulated compositions accurately reflect the primary elemental variability characteristic of lunar regolith. The GCD-2 with Al/Si = 0.34 and Ca/Si = 0.26 served as the baseline group (BG). The various amounts of CaO and Al₂O₃ were added to achieve a compositional gradient. As shown in Table 2, the adjusted mixing proportion covered Al/Si ratios of 0.34–0.58 and Ca/Si ratios of 0.26–0.35, both remaining below the maximum values recorded for authentic lunar regolith (Al/Si ≤ 0.59, Ca/Si ≤ 0.35). After 700 h of thermal curing, all samples were tested using the SHPB. The impact gas pressure of 0.2, 0.3, and 0.4 MPa were applied to investigate the combined effects of Ca/Si ratio, Al/Si ratio, and curing temperature on the dynamic behavior of LRG.

3.4. Analysis and characterization

3.4.1. SHPB test

The SHPB apparatus was employed for dynamic impact testing. As illustrated in Fig. 3(a), the SHPB consisted of a gas-driven projectile launcher, incident and transmission bars, a signal acquisition system, and a signal processing unit. The projectile, incident bar, and transmission bar were fabricated from steel, each with a diameter of 50 mm, density of 7.8 g/cm³, and elastic modulus of 210 GPa. Thus, the longitudinal wave velocity in the bar was determined to be 5190 m/s. Stress and strain responses of the LRG samples were measured using strain gauges mounted on both the incident and transmission bars, in conjunction with the data acquisition system. Due to the relatively low wave impedance of the samples, semiconductor strain gauges (gauge factor = 110 ± 5 %) were used on the transmission bar, while conventional resistance strain gauges (gauge factor = 2.0 ± 5 %) were used on the incident bar. As shown in Fig. 3(b), during testing, the sample was placed between the incident bar and transmission bar, with a thin layer of petroleum jelly applied to both contact surfaces to minimize end friction effects.

To satisfy the stress-uniformity requirement of the SHPB test, the projectile, incident bar, sample, and transmission bar were aligned coaxially before each impact. A copper pulse shaper was inserted to suppress oscillations and refine the incident-pulse rise time, thereby promoting a nearly constant strain-rate loading regime [67]. Raw oscilloscope traces were baseline-corrected and smoothed to obtain pre-processed incident, reflected, and transmitted waves. Using the classical three-wave method [68,69], these signals were converted into stress, strain, and strain-rate. Dynamic stress equilibrium, a prerequisite for data reliability, was evaluated according to one-dimensional stress-wave theory [70]. The summed strain of the incident and reflected waves at the incident bar end was compared with the transmitted-wave strain at the transmission bar end. Close agreement in both amplitude and waveform indicated that the sample experienced uniform axial stress. Impedance mismatch at the bar-sample interface altered stress wave reflection and transmission characteristics, making accurate verification dependent on proper impedance matching [71]. Following this procedure, the three waveforms were time-shifted to a common origin, yielding the different wave curves as shown in Fig. 3(c). The close overlap between the transmitted wave and the superposed incident and reflected wave confirmed that the LRG samples achieved dynamic stress equilibrium during impact loading. Consequently, it has been firmly established that all the test data utilized in this study are accurate and valid.

3.4.2. TGA

Thermogravimetric analysis (TGA) was conducted on the LRG samples by using a NETZSCH STA 449 F3 Jupiter® thermal analyzer under atmospheric conditions. To maintain inert environment and prevent carbonization during thermal decomposition, continuous nitrogen purging was implemented throughout the thermal cycle. The samples underwent controlled heating from ambient temperature (30 ± 2°C) to 1000°C at a constant ramp rate of 10°C/min.

3.4.3. XRD

The mineralogical composition of lunar regolith simulant and LRG samples was analyzed using an X-ray diffractometer (D8 ADVANCE, Bruker) with Cu Kα radiation. All samples were scanned across a 2θ range of 5–80° with a step size of 0.02° and a scanning rate of 2°/min. Prior to analysis, samples underwent sequential processing involving mechanical grinding to achieve particle sizes below 75 μm, followed by vacuum drying at 50°C for 12 h to eliminate moisture-related interference.

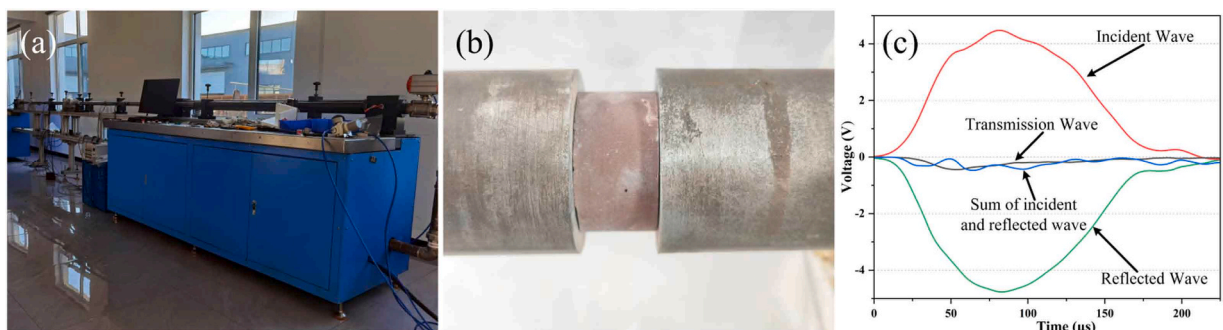


Fig. 3. SHPB test: (a) SHPB apparatus; (b) LRG sample installed in the SHPB; (c) Verification of stress equilibrium.

3.4.4. SEM

The microstructural characterization was performed using a Regulus 8100 field emission scanning electron microscope (FE-SEM, Hitachi High-Tech). Prior to testing, representative fractured sample particles were carefully selected and mounted on aluminum stubs using conductive adhesive under vacuum conditions. To mitigate charging effects during electron beam irradiation, the samples were sputter-coated with a 5 nm thick gold layer using a high-vacuum sputter coater. The prepared samples were then transferred to the SEM chamber and subjected to secondary vacuum to ensure optimal imaging conditions.

3.4.5. Nitrogen adsorption-desorption (NAD) measurements

The porous structure of the LRG was characterized by using NAD measurements (ASAP 2460 analyzer, Micromeritics). Prior to each adsorption measurements, the samples underwent degassing treatment at 40°C under vacuum (1×10^{-3} mbar) for 4 h.

4. Results and discussion

4.1. Dynamic elastic modulus

At the impact gas pressure of 0.3 MPa, the stress–strain responses of the two batches of LRG samples are shown in Fig. 4. Specifically, Fig. 4(a) and (b) corresponded to the LRG samples cured under the first thermal cycle, while Fig. 4(c) and (d) represented the samples subjected to the second thermal cycle. The curves exhibited four stages: elastic (o→a), with a linear rise that defined the dynamic elastic modulus; particle compaction (a→b), with a rapid stress increase as pre-existing microcracks closed, which resulted in increased apparent compressive strength and resistance to deformation; yielding (b→c), with a reduced slope due to the initiation and propagation of new microcracks until the peak; and failure (c→d), with a continuous stress decrease as structural integrity was lost.

The dynamic elastic modulus of the LRG samples was determined based on the slope of the stress–strain curves within the elastic stage (o→a), and the results were summarized in Table 3. The average dynamic elastic modulus of the first batch was 1.78 GPa,

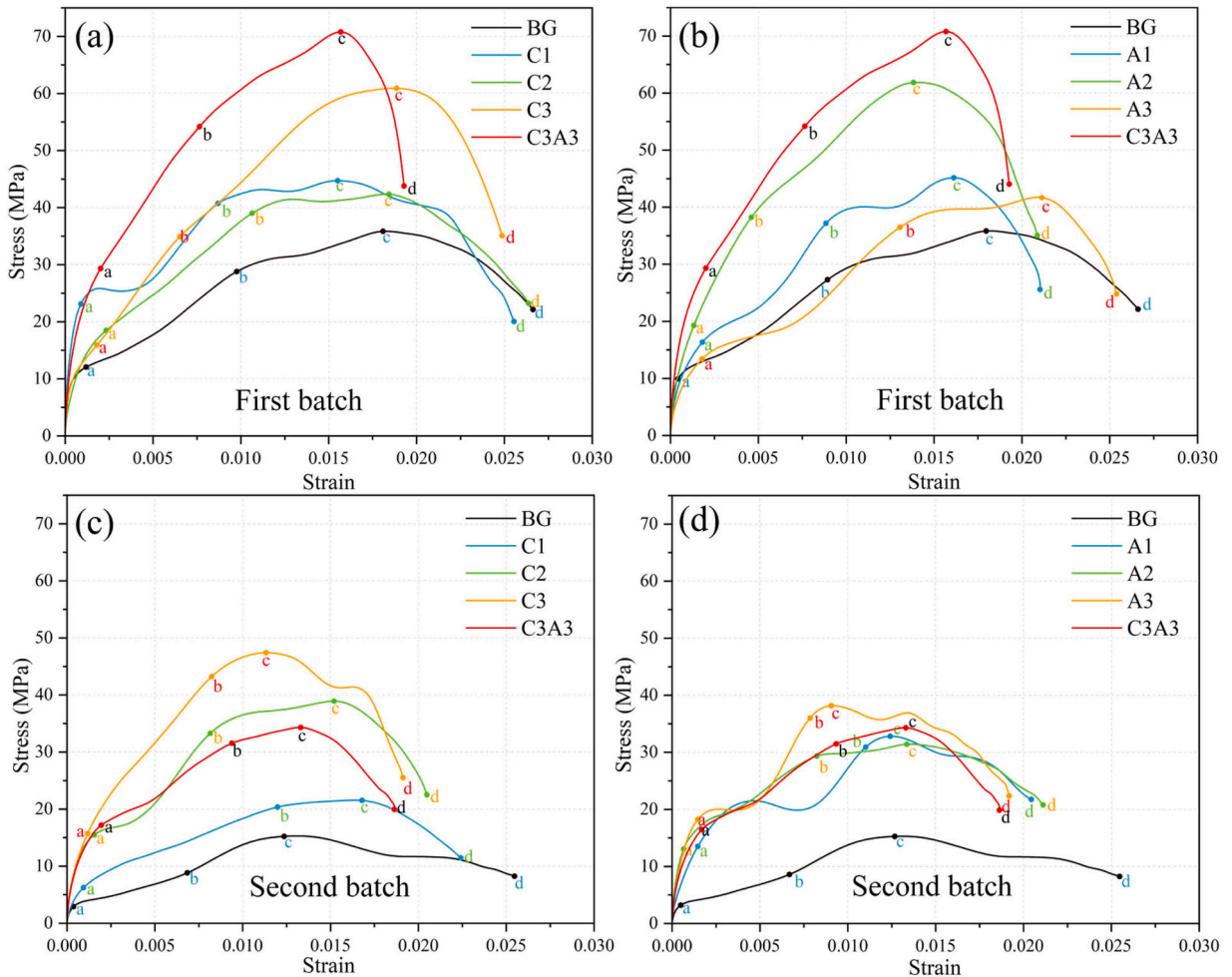


Fig. 4. Stress-strain curves of LRG samples in different curing batches.

Table 3
Dynamic elastic modulus of the LRG samples (GPa).

Curing batches	LRG groups							
	BG	C1	C2	C3	A1	A2	A3	C3A3
First batch	1.54	2.82	1.33	1.83	1.56	1.74	1.07	2.31
Second batch	0.78	0.68	1.05	1.35	0.94	2.38	2.23	1.4

whereas that of the second batch was 1.35 GPa. This notable difference indicated that the samples in the first batch exhibited significantly higher stiffness compared to those in the second batch. This enhancement can be attributed to the more favorable curing conditions of the first batch, which involved a higher curing temperature and an extended curing duration. These conditions promoted more sufficient hydration reactions within the LRG matrix, leading to the formation of abundant cementitious gels. The generated gels effectively bonded unreacted lunar regolith particles and filled internal pores, thereby improving the stiffness and overall integrity of the LRG samples from the first batch. Furthermore, under the first batch's curing condition, the LRG samples with a higher Ca/Si ratio exhibited an average dynamic elastic modulus of 1.99 GPa, while those with a high Al/Si ratio but lower Ca/Si ratio had 1.46 GPa. Such observation highlighted that an increase in Ca/Si ratio contributed to a higher dynamic elastic modulus, implying that Ca-rich in LRG systems were more favorable for enhancing dynamic mechanical properties. Comparative analysis indicated that the groups (C3, A2, and C3A3 groups) exhibiting higher dynamic peak stresses were all cured under the first batch conditions, further confirming that this curing regime was more conducive to the geopolymerization process of LRG. In summary, a comparative evaluation of the dynamic elastic modulus and peak stress between the two batches confirmed that the first curing condition was more favorable for enhancing the overall mechanical properties of the LRG.

4.2. Dynamic compressive strength

The dynamic compressive strength of the LRG samples was determined from the peak stress of the stress–strain curves obtained in the SHPB test [72], and the results are shown in Table 4. From the table, it can be found that as the impact gas pressure increased, the peak compressive strength of the samples increased, demonstrating that the LRG was the strain rate sensitive material. To explain the effects of Ca/Si and Al/Si ratios on dynamic compressive strength, the data acquired at an impact gas pressure of 0.3 MPa were selected. Fig. 5 presents the results for the samples cured under the first batch and tested under impact gas pressure of 0.3 MPa. Compared to the BG group, increasing the Ca/Si ratio in the precursor significantly enhanced the dynamic compressive strength of the LRG. In the C3 group, with a Ca/Si ratio of 0.34, the maximum compressive strength reached 60.92 MPa. This enhancement was attributed to the higher concentration of reactive Ca^{2+} ions introduced by the elevated Ca/Si ratio, which accelerated dissolution of amorphous precursors and promoted the formation of hydration gels [73,74]. The resulting gels effectively densified the LRG matrix.

As shown in Fig. 5, there is a distinct contrast between the effects of Ca/Si and Al/Si ratios on the dynamic compressive strength. Increasing the Ca/Si ratio in the precursor consistently had a positive effect on dynamic compressive strength. In contrast, under low-Ca conditions, increasing the Al/Si ratio initially enhanced the strength, but when the Al/Si ratio exceeded 0.5, the compressive strength of the samples (A3 group) became lower than that of the BG group. It is possible that aluminum acted as a key reactant in the geopolymerization process. The $[\text{AlO}_4]$ tetrahedra bridge silicate chains and construct C-A-S-H and N-A-S-H gel networks, which were responsible for enhancing impact resistance. [75]. However, beyond the optimum Al/Si value (A2 group, Al/Si=0.50), excessive $[\text{AlO}_4]$ units accumulated, sharply increasing the demand for charge-balancing of cations [76]. When the supply of highly polarising Ca^{2+} ions (more effective charge compensators than Na^+ or K^+) was insufficient [77], charge-balancing cannot be maintained and thus

Table 4
Dynamic compressive strength results.

LRG groups	Curing batches	Dynamic compressive strength (MPa)		
		Impact gas pressure 0.2 MPa	Impact gas pressure 0.3 MPa	Impact gas pressure 0.4 MPa
BG	First batch	21.45	35.85	62.13
	Second batch	22.26	15.28	15.07
C1	First batch	32.28	44.72	52.67
	Second batch	10.74	21.56	26.74
C2	First batch	26.44	42.35	51.7
	Second batch	21.42	38.92	42.65
C3	First batch	28.28	60.92	58.93
	Second batch	19.84	25.5	30.47
A1	First batch	25.35	45.16	46.74
	Second batch	15.24	32.82	31.22
A2	First batch	58.69	61.86	59.96
	Second batch	28.93	31.44	33.43
A3	First batch	50.59	31.06	41.72
	Second batch	29.36	38.16	12.85
C3A3	First batch	26.78	70.77	37.86
	Second batch	27.58	34.3	58.63

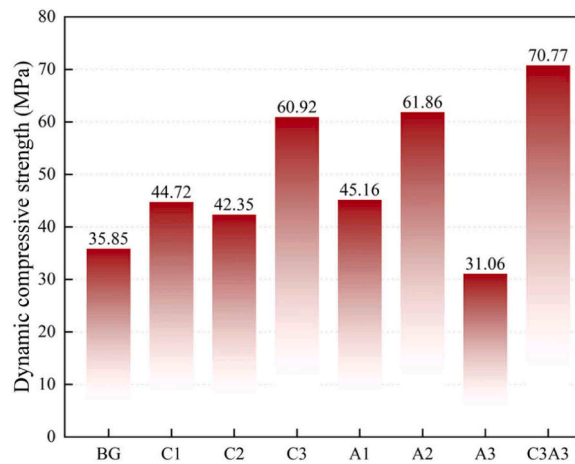


Fig. 5. Dynamic compressive strength of the first batch of LRG with different compositions under impact gas pressure of 0.3 MPa.

the geopolymerization process slowed down [78]. Given the limited curing time of a lunar day phase, this kinetic delay was likely to reduce the formation of gels. Subsequent microstructural analyses would verify this conclusion. By comparison, the C3A3 group exhibited the highest dynamic compressive strength (70.77 MPa). This might be attributed to the pronounced Ca–Al synergistic effect, where the Ca^{2+} not only accelerated the formation of C-A-S-H gels but also mitigated the reduced geopolymerization typically associated with high Al/Si ratios [79]. Hence, such synergistic effect markedly enhanced the dynamic mechanical properties of the LRG.

4.3. Failure pattern

Table 5 illustrates the failure patterns of the samples from the first curing batch under varying impact gas pressures. At impact gas pressure of 0.2 MPa, the fractured samples primarily performed large-sized fragments. As the impact gas pressure increased, the damage severity intensified and the corresponding fragment size noticeably decreased. When the impact gas pressure exceeded 0.3 MPa, the failure of the sample became severe, with the residues mainly consisting of fine particles and powders. A comparative analysis between failure patterns of C3 and A3 groups revealed that under the identical impact gas pressure, the samples with higher Ca/Si ratios produced markedly larger individual fragments than those with elevated Al/Si ratios. Further examination indicated that the samples with higher Ca-rich in precursor yielded more irregular large debris and fewer fine particles. This suggested that increased Ca^{2+} ions content promoted the formation of C-(A)-S-H gels, which effectively bonded the lunar regolith particles and contributed to a denser, more cohesive matrix. Such a compact and uniform matrix structure, with fewer internal pores and microcracks, was more prone to through fracturing under impact loading, thereby generating larger fragments. Additionally, according to fracture mechanics principles, high-strength sample generally exhibited lower fracture energy [80]. As the LRG with elevated Ca/Si ratios possessed higher compressive strength, they were consequently more susceptible to brittle failure under applied stress.

In contrast, the LRG samples with a higher Al/Si ratio fractured into markedly smaller fragments. The higher Al/Si ratio in precursor promoted the formation of additional $[\text{AlO}_4]$ tetrahedra. In the low-Ca system, the shortage of highly polarising Ca^{2+} ions lowered charge-compensation efficiency, slowed geopolymerization, and restricted gel network densification [78]. When Ca^{2+} ions were insufficient to fully balance the negative charge of the aluminosilicate gel, charge compensation must rely on other cations, such as Na^+ . This phenomenon induced the formation of a composite gel system comprising coexisting N-A-S-H and C-A-S-H phases. Due to these two gel phases were thermodynamically and structurally incompatible [81], the matrix became heterogeneous and contained numerous microscale weak interfaces. Under impact loading, cracks nucleated at these defects and propagated along the weak boundaries, causing the samples to disintegrate into finer fragments.

4.4. Energy dissipation

According to one-dimensional stress wave theory [82], and assuming no energy loss during wave propagation in the incident and transmission bars, the energy of the incident stress wave (incident energy) was partially transformed upon reaching the LRG sample. Under such circumstance, A portion of the energy was reflected back into the incident bar (reflected energy), and a portion of the energy was absorbed by the sample (absorbed or dissipated energy). The remainder was transmitted through the specimen into the transmission bar (transmitted energy). The incident energy Q_I , reflected energy Q_R , and transmitted energy Q_T can be calculated according to the principle of energy conservation and stress wave propagation theory, as expressed by the following equations:

Table 5
Failure patterns of the first batch's LRG samples.

LRG groups	Impact gas pressure		
	0.2MPa	0.3MPa	0.4MPa
BG			
C1			
C2			
C3			
A1			
A2			
A3			
C3A3			

$$\begin{cases} Q_I(t) = E_0 C_0 A_0 \int_0^t \varepsilon_I^2(t) dt \\ Q_R(t) = E_0 C_0 A_0 \int_0^t \varepsilon_R^2(t) dt \\ Q_T(t) = E_0 C_0 A_0 \int_0^t \varepsilon_T^2(t) dt \end{cases} \quad (1)$$

$$Q_S(t) = Q_I(t) - Q_R(t) - Q_T(t) \quad (2)$$

where, $Q_I(t)$ represents the incident energy at time t ; $Q_R(t)$ denotes the reflected energy at time t ; $Q_T(t)$ denotes the transmitted energy at time t ; $Q_S(t)$ represents the absorbed energy within the LRG sample, which corresponds to the portion of energy dissipated during impact and is calculated as the difference between the incident, reflected, and transmitted energies; E_0 represents the initial elastic modulus of the specimen; C_0 is its initial longitudinal (elastic) wave velocity; A_0 denotes the cross-sectional area of the specimen; $\varepsilon(t)$ is the axial strain of the specimen at time t , while $\varepsilon_I(t)$, $\varepsilon_R(t)$, and $\varepsilon_T(t)$ represent the strains induced by the incident, reflected, and transmitted stress waves, respectively.

Accordingly, this study employed the energy absorption ratio (defined as the ratio of absorbed energy to incident energy) as a quantitative metric to evaluate the energy dissipation capacity of different LRG formulations [83–85]. Fig. 6 presents the energy absorption ratios of the LRG with different compositions under an impact gas pressure of 0.3 MPa. This ratio was generally positively correlated with the density of the sample. As shown in Fig. 6, compared to the BG group, the LRG samples with higher Ca/Si ratios exhibited significantly increased energy absorption ratios. This observation indicated that increasing the Ca/Si ratio enhanced Ca^{2+} availability in the LRG matrix, which promoted gels formation and improved internal structural integrity, thereby increasing matrix density and energy absorption capacity. In contrast, under low-Ca system, increasing the Al/Si ratio resulted in a non-monotonic trend in energy absorption ratio (initially rising and then declining). This observation suggested that a moderate increase in the Al/Si ratio could optimize the internal structure and improve energy absorption. However, excessive Al^{3+} incorporation might compromise gel network compactness, lower density, and consequently reduce energy dissipation performance. Notably, the C3A3 group exhibited the highest energy absorption ratio, indicating that the synergistic interaction between Ca and Al most effectively improved the matrix structure.

4.5. Dynamic increase factor (DIF)

Based on the above analysis, to further investigate the dynamic mechanical behavior of each sample under varying strain rates, the Dynamic Increase Factor (DIF), which was defined as the ratio of dynamic compressive strength to static compressive strength, can be calculated as follows.

$$DIF = \frac{f_{c,d}}{f_{c,s}} \quad (3)$$

where, $f_{c,d}$ represents the dynamic compressive strength of the LRG samples as listed in Table 4; $f_{c,s}$ denotes the static compressive strength. The measured static compressive strength results for the LRG samples in this study are presented in Table 6.

The average strain rate of the LRG samples during loading was calculated using Eq. (4):

$$\dot{\varepsilon}_s = \frac{1}{t_w - t_0} \int_{t_0}^{t_w} \dot{\varepsilon}(\tau) d\tau \quad (4)$$

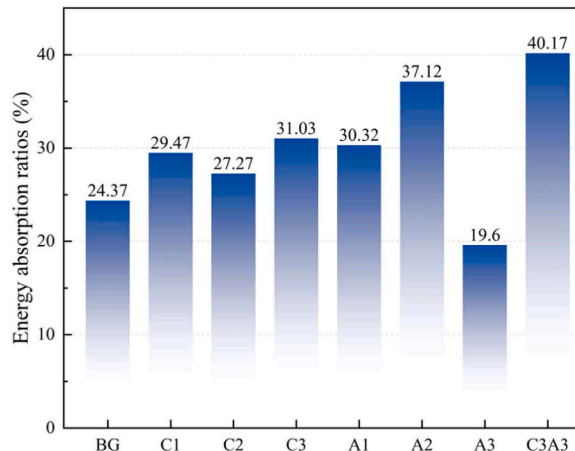


Fig. 6. Energy absorption ratios of LRG with different compositions under impact gas pressure of 0.3 MPa.

Table 6

Static compressive strength of LRG with different compositions (MPa).

LRG groups	BG	C1	C2	C3	A1	A2	A3	C3A3
Static compressive strength	25.1	31.48	33.71	35.28	35.99	33.01	29.86	39.92

in which, $\dot{\epsilon}_s$ denotes the average strain rate of the specimen; t_w and t_0 represent the time duration of the stress wave and the time corresponding to the yield strength on the engineering stress–time curve, respectively; τ denotes the time integration variable.

The fitted relationships between strain rate and DIF for the LRG with different compositions are shown in Fig. 7. As illustrated in Fig. 7(a), compared to the BG group, LRG samples with higher Ca/Si ratios exhibited a pronounced increase in DIF with increasing strain rate, indicating strong strain rate sensitivity. Among them, the C3 group (Ca/Si=0.34) indicated the most significant dynamic strengthening response, suggesting that this composition possessed superior dynamic mechanical properties under high strain rates. Further analysis of the BG group revealed a non-monotonic trend. The DIF increased with strain rate initially, reaching a peak value of 1.54 at 230 s^{-1} , and then began to decline. This behavior might be attributed to microstructural degradation at elevated strain rates.

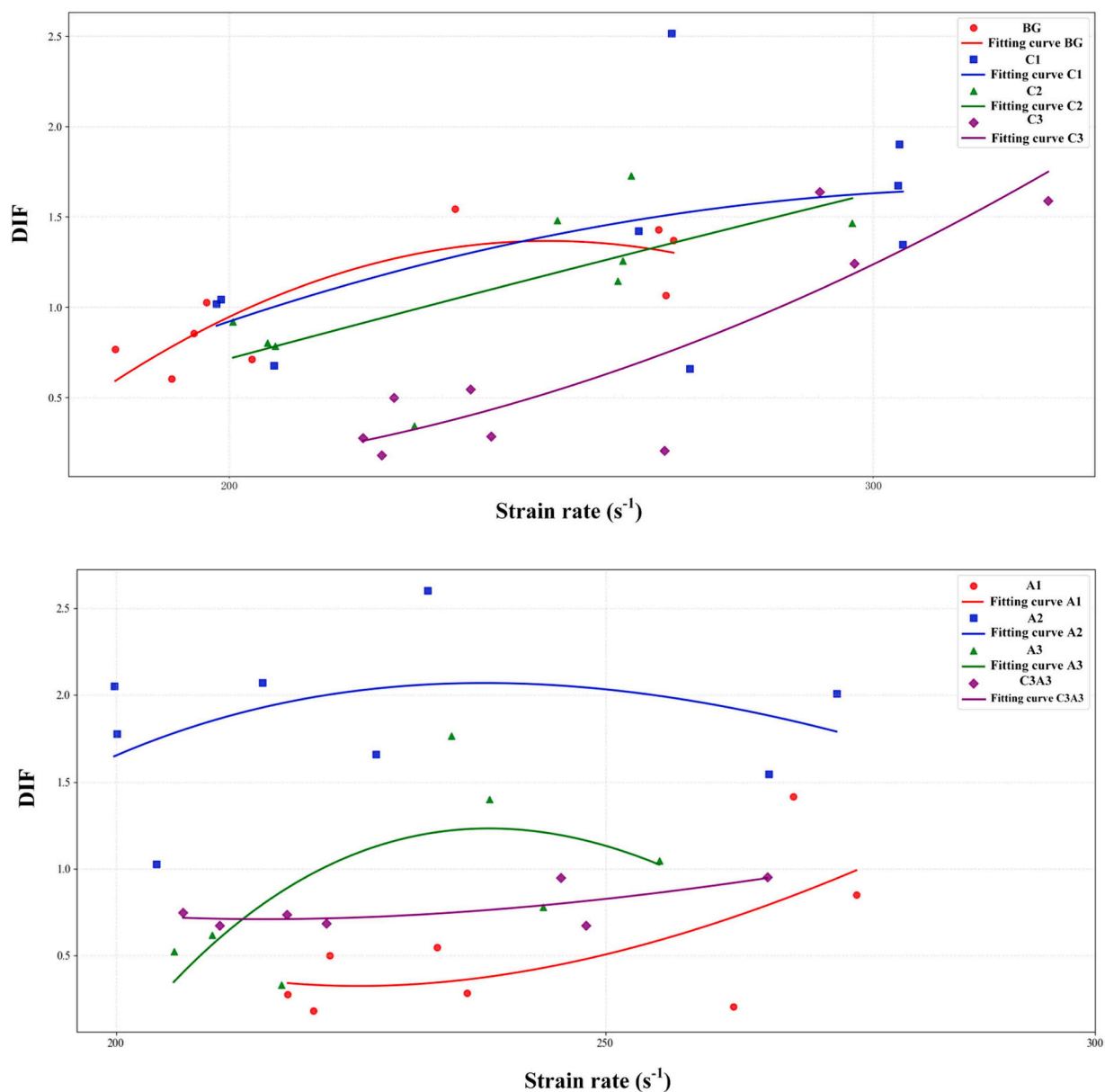


Fig. 7. The fitted relationships between strain rate and DIF for the first batch's LRG with different compositions.

Once the DIF reached its peak, continued strain rate loading likely exacerbated local defects within the LRG matrix, triggering the initiation and propagation of microcracks and pore growth, ultimately resulting in a reduction in dynamic compressive strength.

Fig. 7(b) illustrated that LRG samples with elevated Al/Si ratios exhibited only modest strain rate sensitivity. At same time, the DIF rose only slightly as the strain rate increased. When the Al/Si ratio was 0.51 (A2 group), the DIF gradually leveled off at a relatively high value, indicating that the dynamic compressive strength at high strain rates substantially exceeded the corresponding static strength. By contrast, the A1 group (Al/Si=0.44) exhibited limited strain rate sensitivity, and its DIF remained below 1.0 throughout the entire range. This indicated that the sample had weak impact resistance since the dynamic compressive strength was lower than the corresponding static strength. Further increasing the Al/Si ratio to 0.58 (A3 group) produced the DIF with strain rate trend similar to that of the BG group. These results collectively suggested that a moderate Al/Si ratio enhanced impact performance at high strain rates, whereas excessive Al/Si ratio diminished it.

4.6. Microscopic characterizations

Fig. 8(a) presents the XRD patterns of the unreacted precursor (GCD-2) and the LRG after geopolymerization. The unreacted precursor primarily consisted of crystalline phases such as plagioclase (anorthite and albite), pyroxene, and olivine, with minor amorphous phases. Following alkali activation, the LRG exhibited a broad diffuse halo in the 2θ range of $20\text{--}40^\circ$, with a characteristic peak centered around 30° , indicating the formation of typical amorphous geopolymer gel. Additionally, this diffuse halo showed a systematic shift toward higher diffraction angles, confirming structural reorganization due to the reconfiguration of Si-O-Si and Si-O-Al bond angles during geopolymerization [86]. Simultaneously, a significant reduction in the intensity of diffraction peaks associated with the original crystalline phases (plagioclase and pyroxene) was observed, suggesting the partial dissolution of aluminosilicate

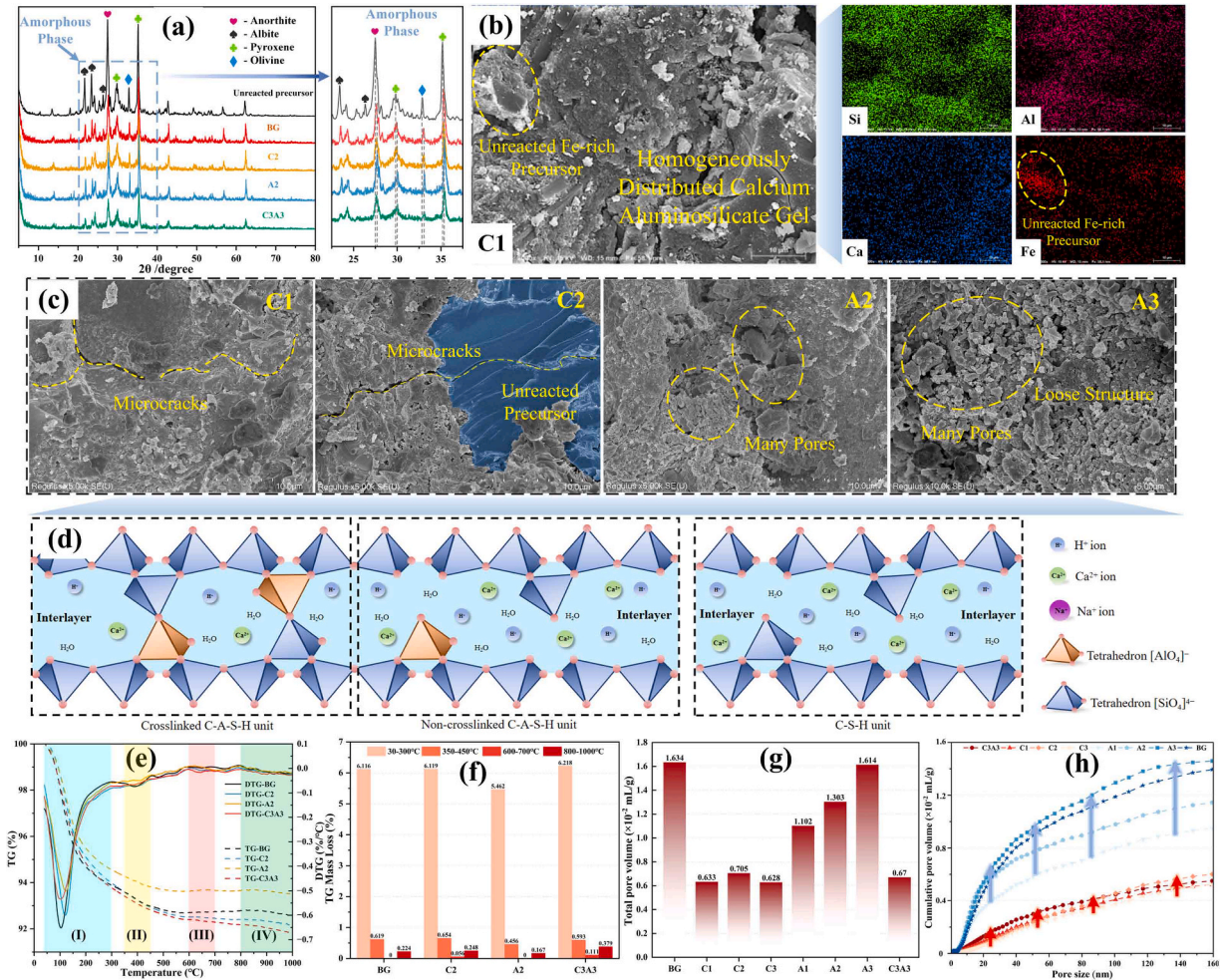


Fig. 8. Microscopic characterizations of the LRG samples with different compositions: (a) XRD spectrograms; (b) The microscopic morphology of element distribution; (c) SEM results; (d) Schematic diagram of the structure of C-(A)-S-H gel; (e) TG-DTG curves; (f) TG mass loss at different stages; (g) Total pore volume results; (h) Cumulative pore volume results.

minerals during the activation process. These results demonstrated that the lunar regolith simulant contained reactive aluminosilicate precursors that could be transformed into amorphous geopolymer gel via alkali activation.

As shown in the SEM morphology of element distribution in Fig. 8(b), the hydration gels were uniformly distributed throughout the LRG matrix and exhibited strong interfacial bonding with the unreacted Fe-rich precursor particles. Fig. 8(c) presents the microstructural morphology of LRG samples with elevated Ca/Si ratios (C1 and C2 groups) and those with higher Al/Si ratios (A2 and A3 groups). Compared to the Al-rich (low Ca/Si) groups, the Ca-rich systems displayed more pronounced microstructural densification. In the Ca-rich samples, a continuous gel phase was observed with reduced microcrack density, forming an integrated matrix that effectively mitigated stress concentrations. This microstructural advantage directly contributed to the improved macroscopic mechanical properties. Notably, as illustrated in Fig. 8(c), the interfacial transition zone (ITZ) between the Ca-rich LRG matrix and unreacted particles showed strong interfacial compatibility, with no evidence of debonding. In contrast, the high Al/Si (low Ca/Si) samples exhibited clear microstructural deficiencies, including increased porosity, elevated microcrack density, reduced hydration degree, and weakened particle bonding. These deficiencies, attributed to poor ITZ integrity and insufficient gel development, ultimately induced a significant decline in the mechanical strength of the LRG system.

During geopolymerization, aluminosilicate precursors were activated in a strongly alkaline environment, where cleavage of Al-O and Si-O bonds initiated step-wise polycondensation into a three-dimensional Si-O-Al network [73]. This process chiefly produced C-(A)-S-H and N-A-S-H gels. Formation of the LRG gel network depended on electrostatic balance between negatively charged aluminosilicate tetrahedra ($[\text{AlO}_4]^-$ and $[\text{SiO}_4]^{4-}$), and compensating cations Ca^{2+} and Na^+ [75]. Previous studies shown that Ca^{2+} surpasses monovalent cations (Na^+ and K^+) in both charge-compensation efficiency and reaction kinetics [87]. Its smaller ionic radius and higher charge density accelerated ion exchange, promoted rapid nucleation of C-(A)-S-H, and established a mechanically robust three-dimensional interpenetrating network. Therefore, increasing the Ca/Si ratio accelerated the formation of C-(A)-S-H while suppressing $[\text{AlO}_4]^-$ tetrahedral cross-linking, thereby inducing the formation of partially Al-free C-S-H phases in Ca-rich systems, as schematically illustrated in Fig. 8(d). These Al-free C-S-H gels, characterised by shorter silicate chains and vacancy-rich interchain configurations [88,89], provided high early strength but also increased brittleness, a result consistent with the conclusions in Section 4.3. In Na-dominated systems (A1–A3 groups), geopolymerization proceeded more slowly. The larger ionic radius and lower charge density of Na^+ , combined with the shortage of Ca^{2+} , further reduced the efficiency of tetrahedral coordination and weakened charge-compensation effectiveness. Consequently, the overall gel formation declined, and the incompatibility of composite gels (C-(A)-S-H and N-A-S-H) in low-calcium systems [81,90] further induced the reduction in mechanical properties. By contrast, the C3A3 group, which combined high Ca/Si and high Al/Si ratios, benefited from an Al-driven kinetic buffering effect that prolonged the polycondensation window [78]. Meanwhile, this synergy, where Ca^{2+} accelerated dissolution and condensation while $[\text{AlO}_4]^-$ units enhanced network cross-linking, provided a tunable pathway for optimizing gel architecture and ultimately enhancing the macroscopic properties of LRG.

Thermogravimetric (TG) and derivative thermogravimetric (DTG) analyses (Fig. 8(e)) were conducted to characterize the thermal evolution of LRG samples within the 30–1000°C range. Based on DTG peak characteristics, the thermal decomposition process was categorized into four distinct stages. Stage I (30–300°C) involved free water desorption (DTG peak at 100–150°C) and the release of interlayer water from the gel network (DTG peak at ~300°C) [55]. Stage II (~400°C) corresponded to the dehydroxylation of calcium hydroxide ($\text{Ca}(\text{OH})_2$) [79,91]. Stage III (~650°C) was associated with the decomposition of calcite (CaCO_3) [79,92]. Stage IV (~900°C) reflected high-temperature phase transformations of hydration products, resulting in the formation of crystalline wollastonite (CaSiO_3) and nepheline ($\text{NaAlSi}_3\text{O}_8$) [93,94].

In Stage I, the observed mass loss (approximately 5.4–6.5 %) was relatively limited, indicating the prior removal of most free water in LRG matrix during lunar daytime thermal exposure. Only trace amounts of structurally bound water remained. The content of interlayer bound water served as a proxy for gels abundance, ranked in descending order as C3A3 > C2 > BG > A2. In Stage II, mass loss variation among formulations revealed compositional dependency. Samples with lower Ca/Si ratios favored the incorporation of Ca^{2+} ions into the hydration gel structure, thereby limiting the formation of free $\text{Ca}(\text{OH})_2$ phases. In Stage III, a minor DTG peak near 650°C was detected exclusively in the C2 and C3A3 groups, attributed to the decomposition of trace CaCO_3 (<0.3 %), indicating slight carbonation in the Ca-rich mixture. This also indicated that Ca-rich systems (compared to low-Ca counterparts) contained a greater amount of free $\text{Ca}(\text{OH})_2$, which subsequently carbonated upon exposure to atmospheric CO_2 , forming calcite. In Stage IV, high-temperature gel to crystal phase transitions resulted in the formation of wollastonite and nepheline. The associated mass loss followed the trend C3A3 > C2 > BG > A2 (Fig. 8(f)), corresponding to the relative abundance of hydration products and aligning with the gel content ranking observed in Stage I. Collectively, these findings further indicated that increasing the Al/Si ratio in low-calcium systems did not enhance gel formation. Rather, it induced an overall reduction in gel quantity. This observation underscored the critical role of Ca-Al composition in controlling both the extent and structural characteristics of the aluminosilicate gel network.

Fig. 8(g) and (h) present the total pore volume and cumulative pore size distribution of LRG samples with different compositions. Increasing the Ca/Si ratio of LRG samples (C1, C2, and C3 groups) compared to the low-Ca systems (A1, A2, and A3 groups) resulted in a systematic reduction in total pore volume, with an average decrease of approximately 63 %. Under low-Ca conditions, a moderate increase in the Al/Si ratio (BG and A1 groups) further diminished pore volume. However, this trend reversed at higher Al/Si ratios. This reversal was attributed to the fact that an excessively high Al/Si ratio in low-Ca systems limited gel network formation during geopolymerization. As a result, the interfacial binding between LRS particles and gel matrix was insufficient, and voids remained partially unfilled, thereby increasing the cumulative pore volume. In contrast, specimens with elevated Ca/Si ratios exhibited markedly reduced cumulative pore volumes and, consequently, higher densification. Specifically, successive increases in Ca/Si lowered the cumulative pore volume by 59.66 %, 55.32 %, and 59.51 % (C1, C2, and C3 groups, compared to the BG group), respectively. This behavior was closely associated with the enhanced formation of C-(A)-S-H gels promoted by higher calcium availability, which

significantly improved microstructural density and ultimately enhanced the dynamic mechanical properties.

Based on the above microstructural analysis, the differences in macroscopic dynamic mechanical properties were mainly controlled by gel type and content, pore structure, and interfacial bonding. The high Ca/Si system significantly reduced porosity and enhanced interfacial bonding by promoting the formation of dense C-(A)-S-H gel, thereby improving dynamic strength. However, the shorter silicon chain structure and aluminum vacancies increased brittleness of the sample. In contrast, the low-Ca/high-Al system showed reduced dynamic strength and a transition of the failure pattern toward ductile behavior, attributable to incompatibility of coexisting gels, high porosity, and weak interfacial bonding. Finally, the synergistic optimization of Ca and Al (C3A3 group) optimized the energy dissipation mechanism under dynamic loading by balancing gel density and network stability.

5. Conclusions

This study elucidated the effects of Ca/Si and Al/Si ratios within the precursor (LRS), as well as curing temperature, on the dynamic mechanical behavior and microstructural characteristics of the LRG. The main conclusions can be summarized as follows:

1. The LRG from the first curing batch (during the lunar day temperature rise phase), exhibited higher dynamic elastic modulus and dynamic compressive strength compared to those from the second batch (during the lunar day cooling phase). This enhancement was attributed to the elevated temperatures during the heating phase, which significantly accelerated geopolymerization, promoted gel formation, and enhanced microstructural densification. These findings suggested that initiating LRG mixing and casting approximately 100 h before the onset of the lunar high temperature window effectively leveraged the thermal environment to accelerate reaction kinetics, thereby improving dynamic mechanical properties.
2. Increasing Ca/Si ratio of precursor to 0.35 simultaneously improved the dynamic elastic modulus, dynamic compressive strength (up to 60.92 MPa), and strain rate sensitivity (as indicated by an increasing DIF with higher strain rates). This improvement was due to the Ca^{2+} ions promoting the dissolution of amorphous phases and the formation of C-(A)-S-H gels, which effectively filled pores and microcracks, thereby enhancing interparticle stress transfer. The resulting densification reduced internal defects and crack initiation sites, leading to the formation of larger fracture fragments under impact and increasing energy absorption capacity. For lunar construction, targeting a Ca/Si ratio near 0.35 could lead to the development of structural units with higher impact resistance.
3. Elevating Al/Si ratio of precursor exhibited a non-linear trend in dynamic mechanical properties, with dynamic modulus, compressive strength, and energy absorption initially increasing, followed by a decline. Meanwhile, strain rate sensitivity decreased with excessive Al/Si ratio. Moderate Al/Si ratio (Al/Si=0.50) facilitated the formation of (C)-N-A-S-H gels via increased $[\text{AlO}_4]$ tetrahedra, improving LRG matrix structural strength. However, excessive Al/Si ratios resulted in over-crosslinked gel networks requiring additional charge-balancing cations. Therefore, the slower reaction kinetics and incompatibility between coexisting gels induced a more porous and weakly bonded matrix, particularly at the ITZ. This increased microstructural defects and promoted stress concentration under impact, shifting the failure pattern toward finer fragment formation. This implied that Al/Si adjustment was essential for developing crack-resistant lunar construction materials.
4. Optimized Ca-Al synergy enhanced overall dynamic mechanical properties. The C3A3 group (Ca/Si=0.35 and Al/Si=0.58) exhibited the highest dynamic compressive strength (70.77 MPa) and significantly improved elastic modulus. This performance was attributable to the synergistic interaction between Ca and Al. Al supplied a robust $[\text{AlO}_4]$ crosslinking network, while Ca ions effectively neutralized the negative charge, mitigating charge imbalance and promoting a high degree of geopolymerization, dense matrix. The resulting microstructure demonstrated superior energy absorption capacity, with impact failure characterized by the formation of large-sized debris, indicative of increased structural integrity. The Ca-Al synergy highlighted the potential for producing high-performance lunar building components that were capable of withstanding dynamic loading.

CRedit authorship contribution statement

Jun Wu: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Zihan Zhou:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Xiyao Zheng:** Writing – review & editing, Validation, Supervision, Project administration, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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